



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

LAB PRACTICAL RECORD

DSA0606: Data Handling and Visualization

Submitted To:

Dr. L. Karthikeyan

Submitted By:

Name: PRIYANKA G

Reg No: 192524124

Department: B.Tech AI & D

INDEX

S.No	Date	Title of Experiment	Page No
1.	11/07/2026	Monthly Sales Data	
2.	11/07/2026	Customer Feedback Analysis	
3.	11/07/2026	Employee Performance Evaluation	
4.	11/07/2026	Product Inventory Management	
5.	14/07/2026	Website Analysis	
6.	14/07/2026	Product Sales Analysis	
7.	14/07/2026	Customer Demographics Analysis	
8.	21/07/2026	Survey Responses Analysis	
9.	21/07/2026	Product Category Analysis	
10.	21/07/2026	Geographic Data Analysis	

DATE: 11/07/2026

SET 1: MONTHLY SALES DATA

PROCEDURE

Step 1: Connect to the Data

- Open Tableau and click Connect → To a File → Text File or Microsoft Excel (depending on your dataset format).
- Select the Monthly Sales data file and click Open.
- Drag the data table onto the canvas, then click the Sheet 1 tab at the bottom.

Step 2: Task 1 – Create a Line Chart (Monthly Sales)

- Drag Month from the Data pane onto the Columns shelf.
- Note: If Month is treated as a string/dimension, ensure it sorts chronologically (Jan, Feb, Mar, etc.).
- Drag Sales (in \$) onto the Rows shelf.
- On the Marks card, change the chart type dropdown from Automatic to Line.
- Right-click the X-axis → Edit Axis → set the title to “Month”.
- Right-click the Y-axis → Edit Axis → set the title to “Sales (\$)”.
- Double-click the sheet title at the top and rename it to “Monthly Sales Trend”.
- Rename the sheet tab at the bottom of the “Monthly Sales Line Chart”. Open a new worksheet.

Step 3: Task 2 – Create a Bar Chart (Top-Selling Products)

- Drag Product Name (or equivalent product field from your full dataset) onto the Columns shelf.
- Drag Sales (or Quantity Sold) onto the Rows shelf.
- On the Marks card, ensure the chart type dropdown is set to Bar.
- Click the Sort Descending icon in the toolbar above the axis, so the top-selling products appear first.
- Drag Sales onto Label on the Marks card to display the exact values on top of the bars.
- Rename the sheet title to “Top-Selling Products” and the sheet tab to “Product Sales Bar Chart”. Open a new worksheet.

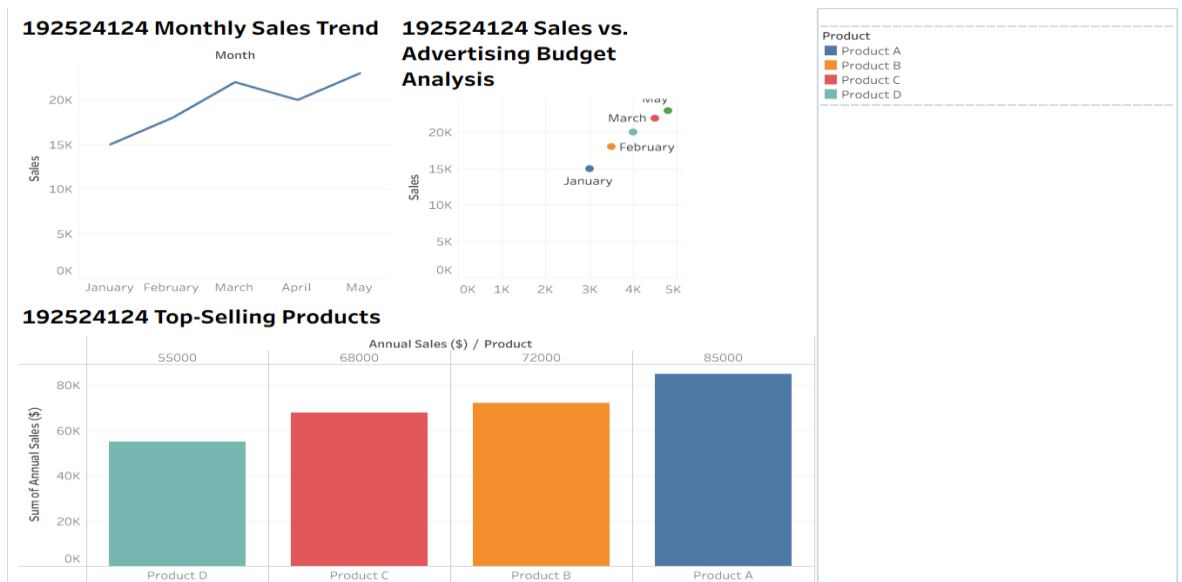
Step 4: Task 3 – Create a Scatter Plot (Budget vs. Sales)

- Drag Advertising Budget (independent variable) onto the Columns shelf.
- Drag Sales (dependent variable) onto the Rows shelf.
- If Tableau aggregates this into a single point, go to the top menu: Analysis → uncheck Aggregate Measures to plot individual data points.
- Alternative: If you want to view it by product or month, drag Product Name or Month to the Detail property on the Marks card.
- Label the axes appropriately (“Advertising Budget (\$)” and “Monthly Sales (\$)”).
- Rename the sheet title to “Sales vs. Advertising Budget Analysis” and the sheet tab to “Budget vs Sales Scatter”. Open a new worksheet.

Step 5: Task 4 – Build an Interactive Dashboard

- Click the New Dashboard icon at the bottom of the screen (the window grid icon).
- From the Dashboard pane on the left, drag “Monthly Sales Line Chart” onto the canvas.
- Drag “Product Sales Bar Chart” and drop it right next to or below the line chart.
- Click on the Line Chart container in the dashboard view, click the small drop-down arrow (or look at the top-right icons of the object border), and select “Use as Filter” (the funnel icon). Do the same for the Bar Chart.
- Adjust sizes, title the dashboard “Monthly Sales Performance Dashboard”, and save your workbook (File → Save).

OUTPUT



DATE: 11/07/2026

SET 2: CUSTOMER FEEDBACK ANALYSIS

PROCEDURE

Step 1: Connect to the Data

- Click the New Data Source icon in the top toolbar or go to Data → New Data Source.
- Connect to your Customer Satisfaction dataset and open a fresh worksheet.

Step 2: Task 1 – Create a Histogram (Age Distribution)

- In the Data pane, right-click the Age measure → Create → Bins...
- Set the size of bins (e.g., intervals of 5 or 10 years depending on preference) and click OK. A new dimension called Age (bins) will appear.
- Drag the newly created Age (bins) onto the Columns shelf.
- Drag the Customer ID measure onto the Rows shelf.
- Right-click the Customer ID pill on the Rows shelf, hover over Measure, and select Count (or Count Distinct). Tableau will automatically render a histogram.
- Label the X-axis “Age Groups” and the Y-axis “Number of Customers”. Title the sheet “Customer Age Distribution”. Open a new worksheet.

Step 3: Task 2 – Create a Pie Chart (Satisfaction Scores)

- Drag Satisfaction Score from the Data pane onto the Columns shelf, and right-click it to change it to a Dimension or Discrete (if it isn’t already), so it treats scores (1–5) as distinct categories.
- Change the Marks card dropdown from Automatic to Pie.
- Drag Satisfaction Score from the Data pane and drop it onto Color on the Marks card.
- Drag Customer ID onto Angle on the Marks card. Right-click it → Measure → Count.
- Drag Customer ID again, this time dropping it onto Label. Change it to Count, right-click it, and select Quick Table Calculation → Percent of Total.
- Rename the sheet title to “Overall Customer Satisfaction Breakdown”. Open a new worksheet.

Step 4: Task 3 – Create a Stacked Bar Chart (Satisfaction by Age Group)

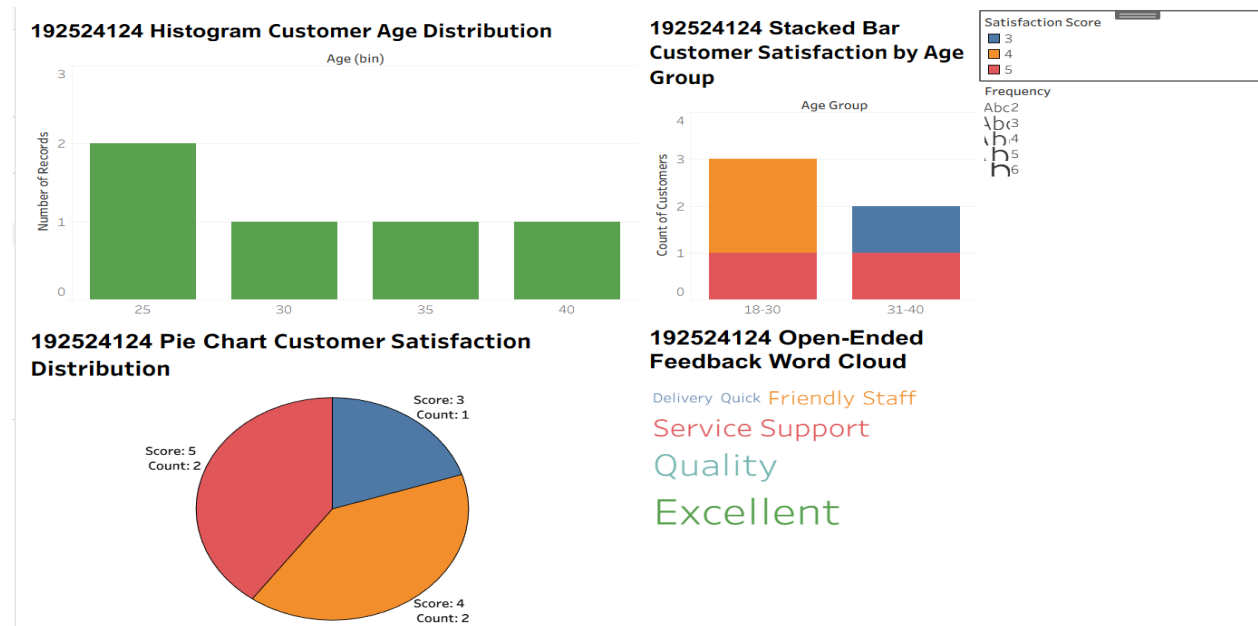
- Drag the Age (bins) dimension (created in Step 2) onto the Columns shelf.
- Drag Customer ID onto the Rows shelf, right-click it → Measure → Count.

- Drag Satisfaction Score and drop it onto Color on the Marks card.
- If the bars are sitting side-by-side instead of stacked, ensure your Marks dropdown is explicitly set to Bar.
- Right-click the Y-axis to verify it is labeled “Count of Customers” and the X-axis is “Age Group”.
- Rename the sheet title to “Customer Satisfaction by Age Group”. Open a new worksheet.

Step 5: Task 4 – Develop a Word Cloud (Open-Ended Feedback)

- Drag the text field containing Open-Ended Customer Feedback onto Text on the Marks card.
- Change the Marks card dropdown from Automatic to Text.
- Drag that same Open-Ended Customer Feedback field (or a Sentiment/Keyword count field if pre-processed) onto Size on the Marks card.
- Right-click the field on the Size property → Measure → Count. You will see the words scale in size based on frequency.
- Drag Satisfaction Score or text frequency onto Color to add an extra layer of visual sentiment analysis.
- Title the sheet “Customer Feedback Sentiment Word Cloud” and save your workbook.

OUTPUT



DATE: 11/07/2026

SET 3: Employee Performance Evaluation

PROCEDURE

Step 1: Connect to the Data

- Open Tableau and click Connect → To a File → Text File / Microsoft Excel.
- Select the Employee Performance data file and click Open.
- Drag the table onto the canvas, then click the Sheet 1 tab at the bottom.

Step 2: Task 1 – Create a Line Chart (Performance Trend Over Time)

- Drag Years of Service (or the specific date/time field from your complete dataset representing time progression) onto the Columns shelf.
- Drag Performance Score onto the Rows shelf.
- On the Marks card, change the chart type dropdown to Line.
- Drag Employee ID onto Color on the Marks card to generate separate lines for individual employee trends and automatically populate the color legend.
- Right-click the X-axis → Edit Axis → set the title to “Years of Service”. Right-click the Y-axis → set the title to “Performance Score”.
- Double-click the sheet title at the top and rename it to “Employee Performance Trend Over Time”. Rename the sheet tab to “Performance Line Chart”. Open a new worksheet.

Step 3: Task 2 – Generate a Bar Chart (Department Distribution)

- Drag Department from the Data pane onto the Columns shelf.
- Drag Employee ID onto the Rows shelf. Right-click the pill → Measure → Count (or Count Distinct).
- Verify the Marks card dropdown is set to Bar.
- Drag the Count (Employee ID) pill from the Rows shelf onto Label on the Marks card to display the count on each bar.
- Title the sheet “Employee Distribution by Department” and rename the tab to “Department Bar Chart”. Open a new worksheet.

Step 4: Task 3 – Build a Scatter Plot (Years of Service vs. Performance)

- Drag Years of Service onto the Columns Shelf.

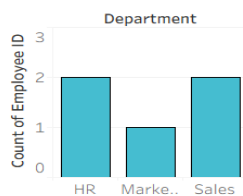
- Drag Performance Score onto the Rows shelf.
- Go to the top menu, select Analysis, and uncheck Aggregate Measures, so individual employee points plot correctly across the grid.
- Drag Employee ID to Detail on the Marks card if you want to inspect specific data points.
- Label the axes as “Years of Service” and “Performance Score”.
- Title the sheet “Correlation: Years of Service vs. Performance Score” and rename the tab to “Service vs Performance Scatter”. Open a new worksheet.

Step 5: Task 4 – Develop a Tableau Dashboard

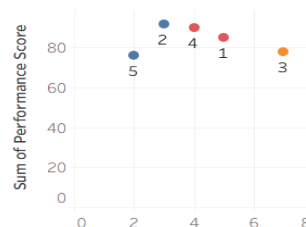
- Click the New Dashboard icon at the bottom of the screen.
- Drag “Performance Line Chart” and “Department Bar Chart” onto the dashboard canvas.
- Select the Bar Chart container, click its drop-down arrow, and check “Use as Filter” so selecting a department dynamically updates the line chart trend for those specific employees.
- Title the dashboard “Employee Performance Evaluation Dashboard” and save the workbook.

OUTPUT

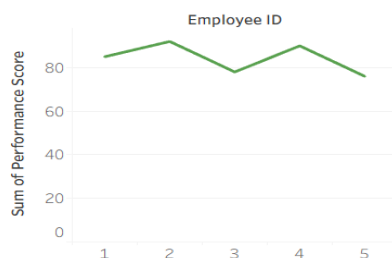
192524124
Employee
Distribution by
Department Bar
Chart



192524124
Correlation: Years of
Service vs.
Performance Score



192524124 Employee
Performance Trend Over
Time



DATE: 11/07/2026

SET 4: Product Inventory Management

PROCEDURE

Step 1: Connect to the Data

Select Data → New Data Source from the top menu bar.

Connect to the Product Inventory data file and open a fresh worksheet.

Step 2: Task 1 – Create a Bar Chart (Product Inventory Quantities)

- Drag Product Name from the Data pane onto the Columns shelf.
- Drag Quantity Available onto the Rows shelf.
- On the Marks card, ensure the chart type dropdown is set to Bar.
- Drag Quantity Available onto Label on the Marks card to display stock levels on top of the bars.
- Right-click the axes to set titles to “Product Name” and “Quantity Available”.
- Title the sheet “Inventory Levels by Product” and rename the tab to “Product Qty Bar Chart”. Open a new worksheet.

Step 3: Task 2 – Generate a Stacked Bar Chart (Quantity within Product Categories)

- Drag Product Category (from your expanded dataset fields) onto the Columns shelf.
- Drag Quantity Available onto the Rows shelf.
- Drag Product Name and drop it onto Color on the Marks card. Ensure the Marks dropdown is explicitly set to Bar to generate the segment stacks.
- Label the axes cleanly (“Product Category” and “Total Quantity Available”).
- Title the sheet “Inventory Breakdown by Category” and rename the tab to “Category Stacked Bar”. Open a new worksheet.

Step 4: Task 3 – Build a Scatter Plot (Price vs. Quantity Available)

- Drag Product Price (from the dataset attributes) onto the Columns shelf.
- Drag Quantity Available onto the Rows shelf.
- Go to the top menu and select Analysis → uncheck Aggregate Measures.
- Drag Product Name onto Detail or Label on the Marks card to identify individual products on the plot.
- Label the axes “Product Price (\$)” and “Quantity Available”.

- Title the sheet “Product Price vs. Quantity Available Analysis” and rename the tab to “Price vs. Qty Scatter”. Open a new worksheet.

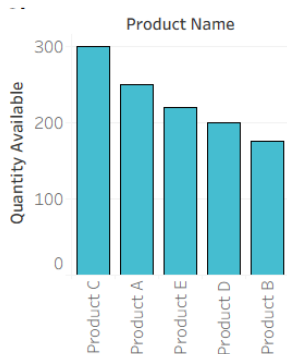
Step 5: Task 4 – Develop a Tableau Dashboard

- Click the New Dashboard icon at the bottom of the tab line.
- Drag both “Product Qty Bar Chart” and “Category Stacked Bar” onto the layout view.
- Enable “Use as Filter” on the stacked bar chart so clicking a specific category instantly focuses the product bar chart on items within that segment.
- Title the dashboard “Product Inventory Management Dashboard” and save your workbook via File → Save.

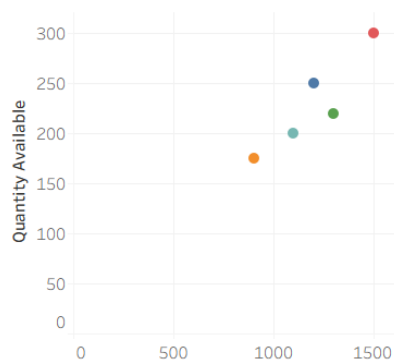
OUTPUT

192524124

Product Qty Bar

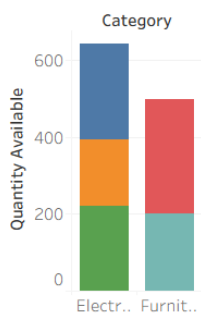


192524124 Price vs Qty Scatter



192524124

Category Stacked Bar



DATE: 14/07/2026

SET 5: WEBSITE ANALYTICS

PROCEDURE

Step 1: Connect to the Data

- Open Tableau and click **Connect** → **To a File** → **Text File / Microsoft Excel**.
- Select your **Website Traffic** data file and click **Open**.
- Drag the data table onto the canvas, then click the **Sheet 1** tab at the bottom.

Step 2: Task 1 – Create a Line Chart (Daily Page Views Trend)

- Drag **Date** from the Data pane onto the **Columns** shelf. *Note: Right-click the Date pill on the Columns shelf and select the continuous day format (e.g., "Day May 8, 2015" in the lower menu section) to get a continuous time series line.*

Note: Right-click the Date pill on the Columns shelf and select the continuous day format (e.g., "Day May 8, 2015" in the lower menu section) to get a continuous time series line.

- Drag Page Views onto the Rows shelf.
- Verify that the Marks card dropdown is set to Line.
- Right-click the X-axis → Edit Axis → set the title to "Date". Right-click the Y-axis → set the title to "Page Views".
- Title the sheet "Daily Page Views Trend" and rename the tab to "Page Views Line Chart". Open a new worksheet.

Step 3: Task 2 – Generate a Bar Chart (Top N Days by Click-Through Rate)

- Drag **Date** onto the **Rows** shelf (right-click it and set it to **Discrete / Exact Date**, so each day is its own row label).
- Drag **Click-through Rate** onto the **Columns** shelf. Ensure the Marks card dropdown is set to **Bar**.
- To create a **Top N** filter: Drag **Date** from the Data pane onto the **Filters** card. In the pop-up window, go to the **Top** tab, select **By field**, choose **Top**, type in your desired number (e.g., 5 or 10), and select **Click-through Rate** via **Sum** or **Average**. Click **OK**.
- Click the **Sort Descending** icon in the toolbar. Drag **Click-through Rate** onto **Label** on the Marks card.
- Title the sheet "**Top Days by Click-Through Rate**" and rename the tab to "**CTR Top N Bar Chart**". Open a new worksheet.

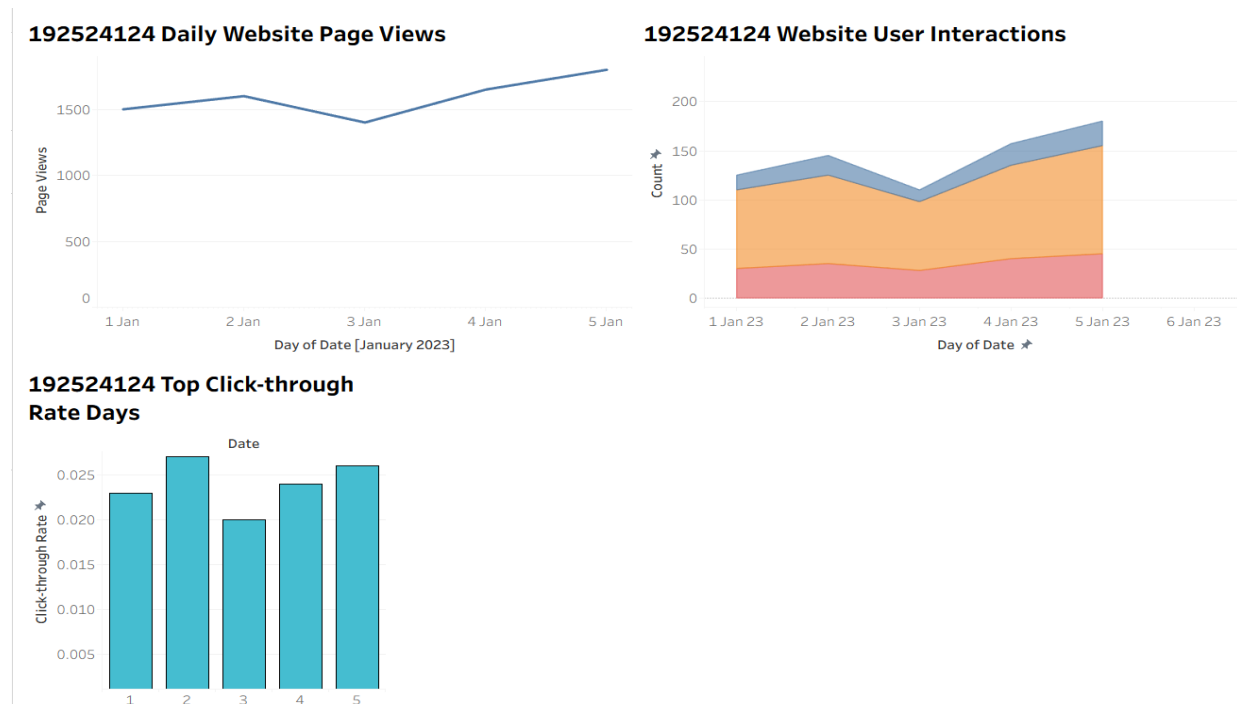
Step 4: Task 3 – Develop a Stacked Area Chart (User Interaction Distribution)

- Drag **Date** (continuous day) onto the **Columns** shelf.
- Drag **Measure Values** from the Data pane onto the **Rows** shelf.
- Find the **Measure Names** filter card that automatically generates, edit it, and check *only* the specific interaction metrics: **Likes**, **Shares**, and **Comments**.
- Drag **Measure Names** from the Data pane and drop it onto **Color** on the Marks card.
- Change the Marks card dropdown from *Automatic* to **Area**.
- Title the sheet "**Distribution of User Interactions over Time**" and rename the tab to "**Interactions Stacked Area**". Open a new worksheet.

Step 5: Task 4 – Build a Tableau Dashboard

- Click the **New Dashboard** icon at the bottom.
- Drag "**Page Views Line Chart**" and "**CTR Top N Bar Chart**" onto the dashboard.
- Select the Line Chart, click on its drop-down arrow, and check "**Use as Filter**" to let users filter the top CTR bar chart by date ranges.
- Title the dashboard "**Website Traffic & Analytics Dashboard**" and save.

OUTPUT



DATE: 14/07/2026

SET 6: PRODUCT SALES ANALYSIS

PROCEDURE

Step 1: Connect to the Data

- Select **Data** → **New Data Source** and connect to your **Monthly Product Sales** dataset. Open a new worksheet.
- *Data Prep Note:* If your fields are separate columns (January Sales, February Sales, March Sales), select those three columns in Tableau's Data Source screen, right-click, and choose **Pivot**. This turns columns into **Pivot Field Names** (Months) and **Pivot Field Values** (Sales).

Step 2: Task 1 – Create a Grouped Bar Chart (Q1 Product Sales)

- Drag **Pivot Field Names (Month)** onto the **Columns** shelf.
- Drag **Product Name** onto the **Columns** shelf right next to Month.
- Drag **Pivot Field Values (Sales)** onto the **Rows** shelf.
- Ensure the Marks card dropdown is set to **Bar**.
- Drag **Product Name** onto **Color** on the Marks card to separate the bars visually within each month's cluster.
- Label the axes "**Q1 Months**" and "**Sales (\$)**". Title the sheet "**Product Sales Comparison: Q1**" and rename the tab to "**Q1 Grouped Bar**". Open a new worksheet.

Step 3: Task 2 – Generate a Stacked Area Chart (Overall Sales Trend)

- Drag **Pivot Field Names (Month)** onto the **Columns** shelf.
- Drag **Pivot Field Values (Sales)** onto the **Rows** shelf.
- Drag **Product Name** onto **Color** on the Marks card.
- Change the Marks card dropdown selector to **Area**.
- Title the sheet "**Overall Product Sales Trend**" and rename the tab to "**Total Sales Stacked Area**". Open a new worksheet.

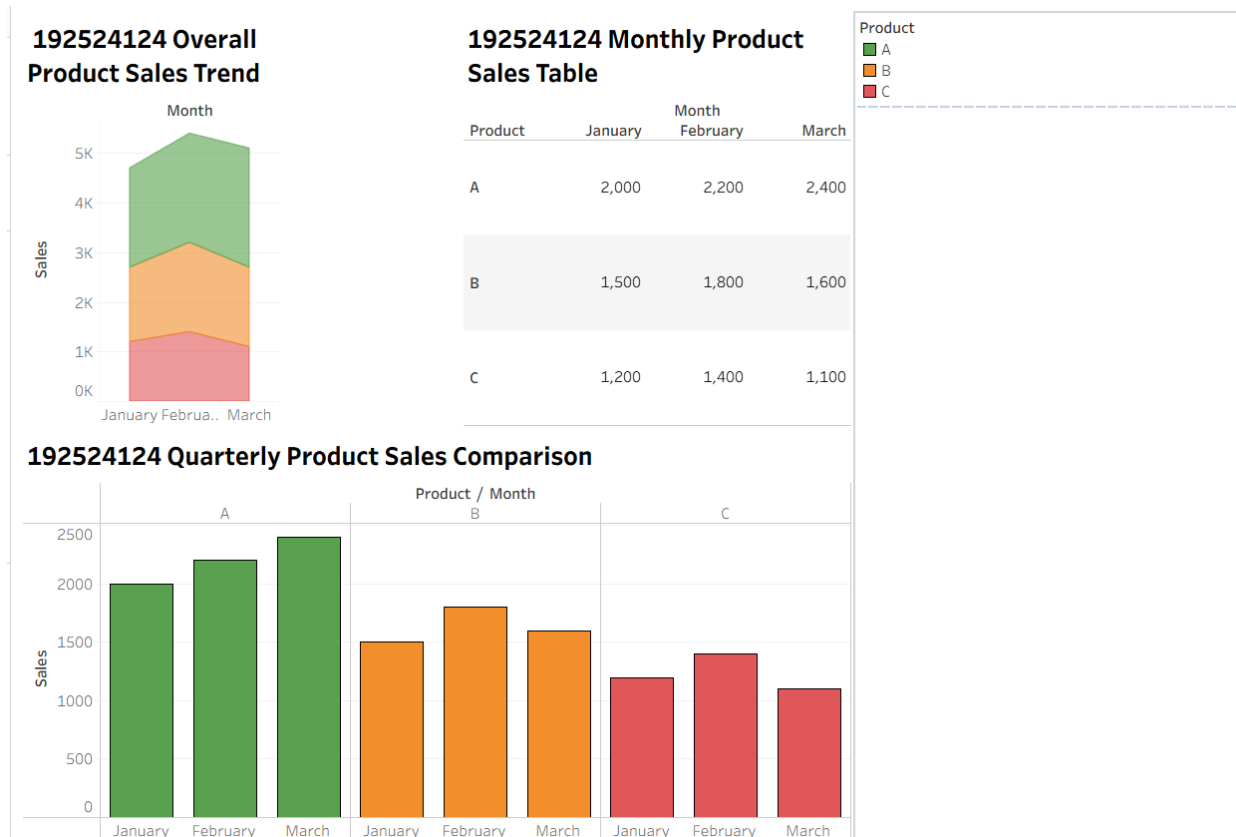
Step 4: Task 3 – Build a Table (Monthly Sales Data Matrix)

- Drag **Product Name** onto the **Rows** shelf.
- Drag **Pivot Field Names (Month)** onto the **Columns** shelf.
- Drag **Pivot Field Values (Sales)** onto **Text** on the Marks card.
- Title the sheet "**Q1 Product Sales Data Matrix**" and rename the tab to "**Sales Summary Table**". Open a new worksheet.

Step 5: Task 4 – Develop a Tableau Dashboard

- Click the **New Dashboard** icon.
- Drag "**Q1 Grouped Bar**", "**Total Sales Stacked Area**", and "**Sales Summary Table**" onto the layout view grid.
- Activate "**Use as Filter**" on the table view so clicking a single cell or product row highlights/filters both visual trends.
- Title the dashboard "**Q1 Product Sales Performance Analysis**" and save.

OUTPUT



DATE: 14/07/2026

SET 7: CUSTOMER DEMOGRAPHICS ANALYSIS

PROCEDURE

Step 1: Connect to the Data

- Select Data → New Data Source, connect to your Customer Demographics dataset, and open a new worksheet.

Step 2: Task 1 – Create a Bar Chart (Age Distribution)

- Right-click Age in the Data pane → Create → Bins... Set the bin size (e.g., 5 or 10 years) and click OK.
- Drag Age (bins) onto the Columns shelf.
- Drag Customer ID onto the Rows shelf. Right-click the pill → Measure → Count.
- Set the Marks dropdown to Bar. Label the axes "Age Range" and "Number of Customers".
- Title the sheet "Customer Age Demographics" and rename the tab to "Age Bar Chart". Open a new worksheet.

Step 3: Task 2 – Generate a Pie Chart (Gender Distribution)

- Change the Marks card dropdown from Automatic to Pie.
- Drag Gender from the Data pane onto Color on the Marks card.
- Drag Customer ID onto Angle on the Marks card. Right-click it → Measure → Count.
- Drag Gender and Customer ID onto Label on the Marks card. Right-click the Customer ID label pill → Quick Table Calculation → Percent of Total.
- Title the sheet "Customer Breakdown by Gender" and rename the tab to "Gender Pie Chart". Open a new worksheet.

Step 4: Task 3 – Build a Table (Demographic Profile Ledger)

- Drag Customer ID onto the Rows shelf. (Right-click and ensure it is set to discrete dimension if needed.)
- Drag Age, Gender, and Income onto the Rows' shelf to the right of Customer ID, so they line up as a clean table layout.
- Optional formatting: To clean up empty text placeholder flags, drag Customer ID (set to Count) onto Text on the Marks card, or just leave text fields blank.
- Title the sheet "Customer Demographics Master Record" and rename the tab to "Demographics Table". Open a new worksheet.

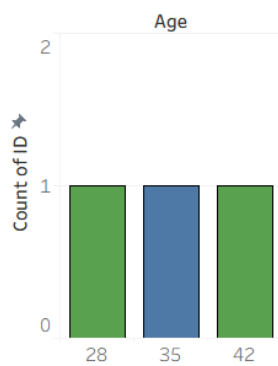
Step 5: Task 4 – Develop a Tableau Dashboard

- Click the New Dashboard icon at the bottom panel interface.
- Drag "Age Bar Chart", "Gender Pie Chart", and "Demographics Table" onto the dashboard canvas grid workspace.

- Turn on "Use as Filter" on the Gender Pie Chart, so clicking a specific slice narrows the table records instantly to that specific segment.
- Title the dashboard "Customer Demographics Profile Hub" and save via File → Save.

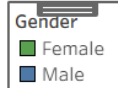
OUTPUT

**192524124
Customer Age
Distribution**

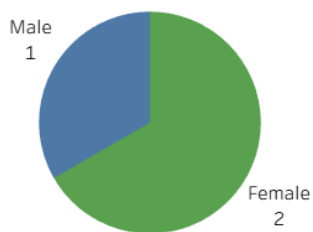


**192524124 Customer
Demographics Table**

ID	Age	Gender	Income	
1	28	Female	50000	A..
2	35	Male	60000	A..
3	42	Female	75000	A..



192524124 Customer Gender Distribution



DATE: 21/07/2026

SET 8: SURVEY RESPONSES ANALYSIS

PROCEDURE

Step 1: Connect to the Data

- Open Tableau and click Connect → To a File → Text File / Microsoft Excel.
- Select your Survey Responses dataset and click Open.
- Drag the table onto the canvas, then click the Sheet 1 tab at the bottom.

Step 2: Task 1 – Create a Grouped Bar Chart (Distribution of Answers for Question 1)

- Drag Question 1 from the Data pane onto the Columns shelf.
- Drag Survey ID onto the Rows shelf. Right-click the pill → Measure → Count.
- On the Marks card, set the chart type to Bar.
- Drag Question 1 onto Color on the Marks card to distinguish answer choices visually.
- Drag Count (Survey ID) onto Label on the Marks card to display total response counts on top of each bar.
- Right-click axes to edit titles (“Question 1 Response Options” and “Number of Responses”).
- Title the sheet “Question 1 Response Distribution” and rename the tab to “Q1 Grouped Bar”. Open a new worksheet.

Step 3: Task 2 – Generate a Stacked Bar Chart

- Data Prep Note: In Tableau’s Data Source tab, select columns from Question 1, Question 2, and Question 3, right-click, and click Pivot. This creates Pivot Field Names (Question Number) and Pivot Field Values (Answers A, B, C, D).
- Drag Pivot Field Names onto the Columns shelf.
- Drag Survey ID onto the Rows shelf → Right-click → Measure → Count.
- Drag Pivot Field Values onto Color on the Marks card.
- Set the Marks dropdown selector explicitly to Bar to stack the response options within each question bar.
- Label the axes (“Survey Questions” and “Total Response Count”).
- Title the sheet “Overall Response Distribution (Q1 – Q3)” and rename the tab to “Questions Stacked Bar”. Open a new worksheet.

Step 4: Task 3 – Build a Table (Survey Response Summary)

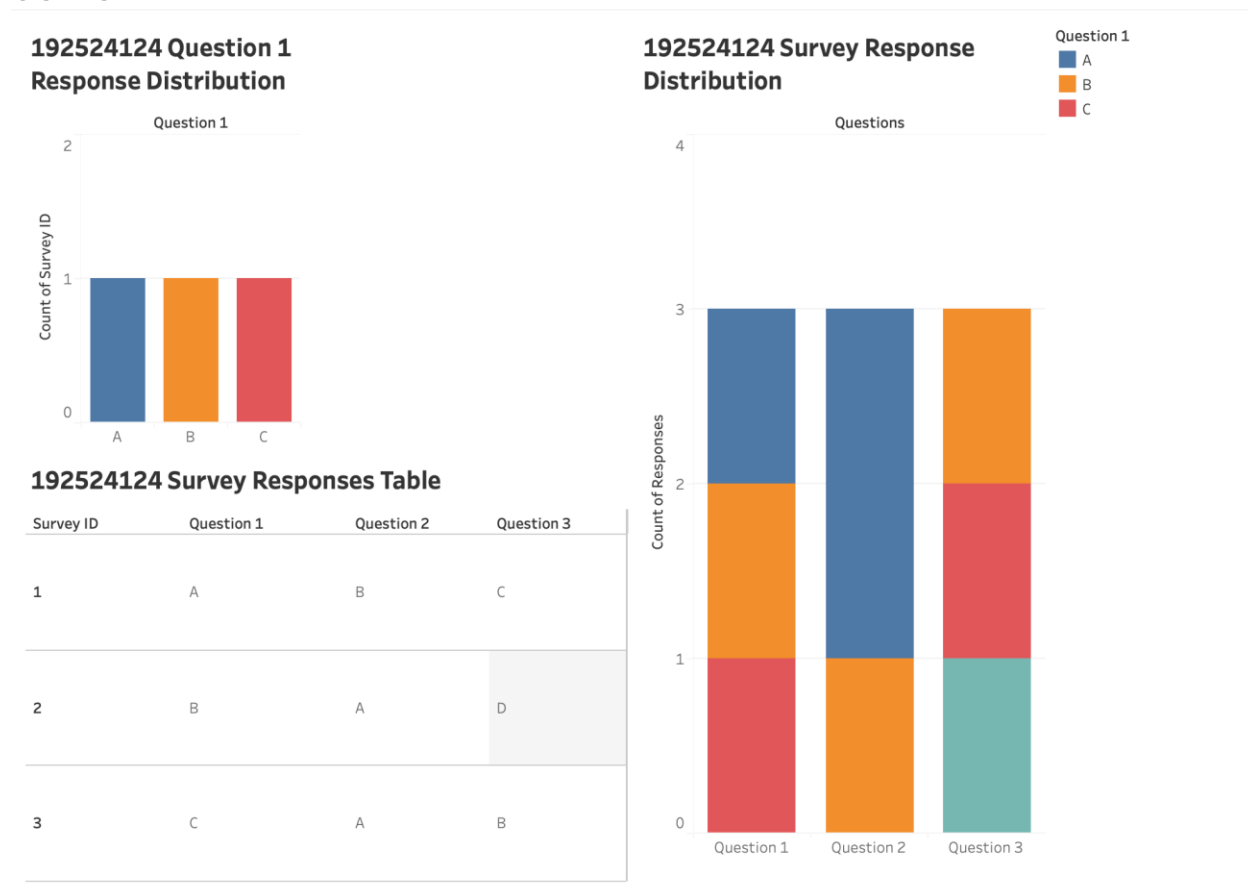
- Drag Survey ID onto the Rows shelf.

- Drag Question 1, Question 2, and Question 3 onto the Rows shelf to the right of Survey ID.
- Right-click the header areas if needed to format fonts, column boundaries, and titles.
- Title the sheet “Survey Response Ledger” and rename the tab to “Survey Data Table”.
Open a new worksheet.

Step 5: Task 4 – Develop a Tableau Dashboard

- Click the New Dashboard icon at the bottom of the tab line.
- Drag “Q1 Grouped Bar”, “Questions Stacked Bar”, and “Survey Data Table” onto the dashboard canvas.
- Click the drop-down menu on the Questions Stacked Bar container and select “Use as Filter” so selecting an answer category filters the detailed response table below.
- Title the dashboard “Survey Responses Interactive Dashboard” and save your workbook.

OUTPUT



DATE: 21/07/2026

SET 9: PRODUCT CATEGORY ANALYSIS

PROCEDURE

Step 1: Connect to the Data

Select Data → New Data Source, connect to your Product Categories dataset, and open a new worksheet.

Step 2: Task 1 – Generate a Funnel Chart (Sales Conversion Process)

- Drag Sales (in \$) onto the Columns Shelf.
- Drag Category onto the Rows shelf.
- Click the Sort Descending button on the top toolbar so categories order from largest sales amount to smallest.
- On the Marks card, change the chart type to Automatic or Bar (or Area for a continuous tapered appearance).
- Drag Category onto Color on the Marks card.
- Drag Sales (in \$) onto Label on the Marks card.
- Title the sheet "Sales Conversion Funnel by Category" and rename the tab to "Category Funnel Chart". Open a new worksheet.

Step 3: Task 2 – Build a Table (Category Sales Data)

- Drag Category onto the Rows shelf.
- Drag Sales (in \$) onto Text on the Marks card.
- Double-click the header to rename columns cleanly.
- Title the sheet "Product Category Sales Summary Table" and rename the tab to "Category Sales Table". Open a new worksheet.

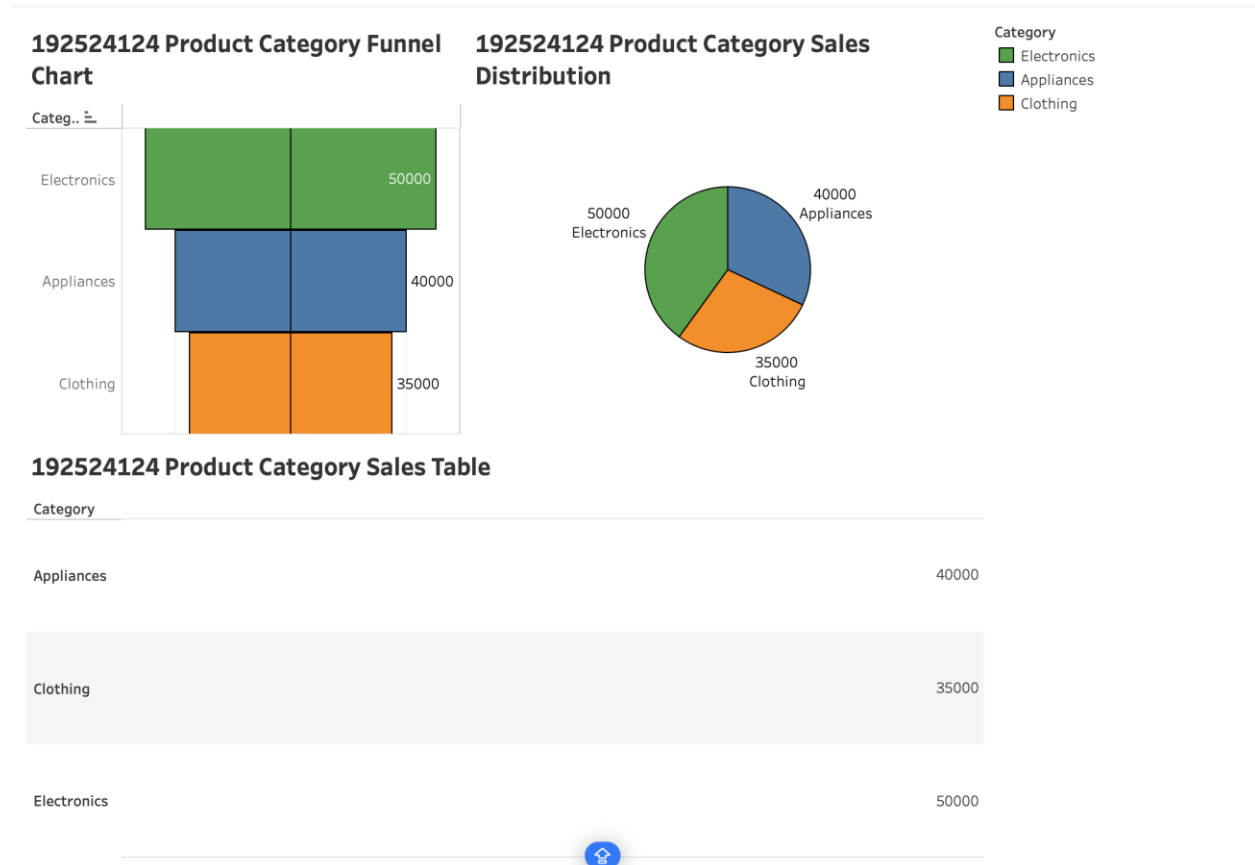
Step 4: Task 4 – Create a Pie Chart (Sales Distribution Across Categories)

- On a new worksheet, change the Marks card dropdown from Automatic to Pie.
- Drag Category onto Color on the Marks card.
- Drag Sales (in \$) onto Angle on the Marks card.
- Drag Category and Sales (in \$) onto Label on the Marks card. Right-click the Sales label pill → Quick Table Calculation → Percent of Total.
- Title the sheet "Sales Share by Category" and rename the tab to "Category Sales Pie Chart". Open a new worksheet.

Step 5: Task 3 – Develop a Tableau Dashboard

- Click the New Dashboard icon.
- Drag "Category Sales Pie Chart", "Category Funnel Chart", and "Category Sales Table" onto the layout of workspace canvas.
- Enable "Use as Filter" on the Pie Chart view so selecting a sector dynamically highlights data in the funnel and table.
- Title the dashboard "Product Category Performance Dashboard" and save via File → Save.

OUTPUT



DATE: 21/07/2026

SET 10: GEOGRAPHIC DATA ANALYSIS

PROCEDURE

Step 1: Connect to the Data

- Select Data → New Data Source, connect to your Geographic Data file, and open a new worksheet.

Step 2: Task 1 – Create a Map Chart (Distribution of Cities)

- In the Data pane, ensure City has a Geographic Role assigned (Right-click City → Geographic Role → City).
- Double-click City (or drag City to the canvas center). Tableau will automatically place Longitude on Columns and Latitude on Rows to generate a map view.
- Drag Population onto Size on the Marks card to scale mark sizing geographically.
- Drag Avg. Temperature or City onto Color on the Marks card.
- Title the sheet “Geographic Distribution of Cities” and rename the tab to “City Map Chart”. Open a new worksheet.

Step 3: Task 2 – Generate a Scatter Plot (Temperature vs. Population Size)

- Drag Avg. Temperature onto the Columns shelf.
- Drag Population onto the Rows shelf.
- From the top menu, click Analysis → uncheck Aggregate Measures.
- Drag City onto Detail and Label on the Marks card, so each point identifies a city.
- Label the X-axis “Average Temperature (°F)” and Y-axis “Population Size”.
- Title the sheet “Average Temperature vs. Population Scatter” and rename the tab to “Temp vs Population Scatter”. Open a new worksheet.

Step 4: Task 3 – Build a Table (Geographic Summary Data)

- Drag City onto the Rows shelf.
- Drag Population, Avg. Temperature, and Elevation onto the Rows shelf right next to City.
- Format header columns for clarity.
- Title the sheet “City Geographic Metrics Summary” and rename the tab to “Geographic Data Table”. Open a new worksheet.
- Step 5: Task 4 – Develop a Tableau Dashboard
- Click the New Dashboard icon at the bottom.
- Drag “City Map Chart”, “Temp vs Population Scatter”, and “Geographic Data Table” onto the dashboard layout pane.

- Click the drop-down menu on the City Map Chart and check “Use as Filter”.
- Title the dashboard “Geographic Data Interactive Dashboard” and save your workbook via File → Save.

OUTPUT

